

RESEARCH ARTICLE

Fluorine-Aligned Electrostatic Gatekeepers Regulating Polysulfide Migration for Stable Lithium–Sulfur Batteries

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ABSTRACT

The commercial viability of lithium–sulfur batteries is fundamentally limited by the shuttle effect, in which the uncontrolled migration of soluble lithium polysulfides triggers cascading active material loss and catastrophic anode degradation. While conventional separator engineering strategies predominantly rely on affinity-based adsorption, these approaches inevitably compromise Li⁺ transport and lead to irreversible sulfur loss. Herein, we report a paradigm-shifting polysulfide-repulsive interface, featuring highly oriented C–F dipoles anchored on an MXene scaffold. Unlike passive adsorption barriers, this electrostatic gatekeeper exploits the strong molecular polarization of β -phase poly(vinylidene fluoride) (β -PVdF) to create a persistent electrostatic field that actively repels anionic polysulfide species. This push-back mechanism effectively confines redox intermediates within the cathode region. Therefore, the electrostatically repulsive separator enables an exceptional cycling lifespan (804 mAh g^{−1} after 700 cycles at 0.5C). This work establishes a dipole-engineered transport regulation strategy that provides a robust blueprint for achieving long-term interfacial stability in high-energy-density storage systems.

1 | Introduction

Lithium–sulfur (Li–S) batteries are among the most promising candidates for next-generation energy storage systems, combining an exceptionally high theoretical energy density (2600 Wh kg^{−1}) with the natural abundance and environmental sustainability of sulfur [1, 2]. Despite these advantages, their practical implementation remains severely limited by rapid capacity decay and low sulfur utilization [3, 4]. These challenges stem from the inherently multistep redox chemistry of sulfur, which generates soluble lithium polysulfides (LiPSs) that shuttle between the cathode and Li-metal anode [5, 6]. The uncontrolled crossover of

LiPSs triggers persistent parasitic reactions, leading to continuous depletion of active sulfur species and accelerated anode corrosion [7, 8].

A wide range of cathode- and anode-oriented strategies have been explored to address these issues. In particular, extensive efforts have focused on cathode-side sulfur host engineering to immobilize dissolved LiPSs [9, 10], as well as on the development of protective interfacial layers to mitigate side reactions at the metal anode [11–15]. However, the dissolution of LiPSs is thermodynamically favorable and thus cannot be completely eliminated; similarly, concentration-gradient-driven crossover is intrinsically

unavoidable during cell operation [16]. As a result, degradation processes inevitably occur across multiple interfaces, ultimately leading to irreversible cell deterioration.

One of the most effective approaches for preventing dissolution of LiPSs is to modify the separator to serve as the final defensive barrier against polysulfide transport, owing to its ability to directly regulate polysulfide migration within the cell [17]. Hence, separator engineering has emerged as a compelling strategy in this area. Most strategies reported to date rely on adsorption-driven mechanisms to regulate LiPSs migration through physical or chemical confinement [18–20]. While these approaches are conceptually simple and can effectively suppress polysulfide migration, adsorption-based strategies are generally effective under moderate sulfur loadings; however, under practical high-loading conditions, their performance becomes fundamentally limited by the finite density of available adsorption sites. Continuous accumulation of LiPSs within the separator pore ultimately leads to saturation, and once the adsorption threshold is exceeded, the suppression effect is inevitably lost. Moreover, irreversible loss of active material due to immobilization on the separator surface inevitably results in a continuous decrease in Coulombic efficiency (CE).

In this work, we shift the paradigm from passive capture to active rejection. We introduce a new separator design featuring a polysulfide-repulsive interface. A polysulfide-phobic separator is constructed by anchoring F-aligned β -phase poly(vinylidene fluoride) (β -PVdF) chains onto an MXene scaffold. This highly polarized interface generates electrostatic repulsion against negatively charged LiPSs species, effectively preventing their approach to the separator. In principle, this repulsion-based approach can function effectively regardless of LiPSs concentration and sustain its regulatory function over extended cycling, making it fundamentally more robust than conventional adsorption-driven mechanisms. Moreover, this architecture can be fabricated via a simple doctor-blade process without complex multistep treatments, offering a practical and scalable route toward high-loading Li–S batteries.

2 | Results and Discussion

The architectural contrast between a commercial polypropylene (PP) separator and our electrostatically repulsive separator (ER-PP) is illustrated in Figure 1a. The pristine PP separator, being nonpolar and chemically inert, lacks the functional interactions required to regulate LiPS species. Therefore, LiPSs readily permeate through the porous membrane, triggering uncontrolled shuttle processes and severe interfacial degradation of the Li-metal anode. By redirecting LiPSs back to the cathode, the intrinsic limitations of conventional separators can be fundamentally mitigated. In this study, we propose an electrostatic repulsion strategy designed to actively push back polysulfides, as illustrated in the bottom scheme of Figure 1a. To realize this concept, we developed a functional separator coated with an F-aligned β -PVdF-anchored MXene scaffold, creating a highly polarized interfacial environment. In this architecture, the aligned C–F dipoles of β -PVdF, together with the negatively terminated MXene sheets, cooperatively generate a strong electrostatic repulsion against

LiPSs species, effectively pushing them back toward the cathode (Figure 1b).

To construct the LiPSs-repulsive interface, the coating material must possess exceptionally strong polarity and dielectric permittivity, enabling the formation of a robust electrostatic barrier against anionic polysulfide species. Among potential candidates, β -PVdF is particularly attractive because its molecular dipoles ($-\text{CH}_2-\text{CF}_2-$) are aligned in the parallel configuration, yielding a remarkably high net dipole moment ($\sim 8 \times 10^{-30}$ C·m) (Figure 1c) [21]. In contrast, α -PVdF adopts a trans–gauche conformation in which individual dipoles largely cancel each other out, resulting in negligible macroscopic polarization. However, because dipole alignment in bulk PVdF is often incomplete, a mechanically robust and chemically active scaffold [22], namely MXene, is introduced as an interfacial framework to stabilize the β -phase conformation through interfacial coupling (Figure 1c).

$\text{Ti}_3\text{C}_2\text{T}_x$ MXene was specifically selected based on three critical factors. First, it provides strong interfacial coupling with β -PVdF terminations: MXene surfaces intrinsically exhibit high electronegativity owing to their $-\text{F}$, $-\text{O}$, and $-\text{OH}$ terminations, which effectively accommodate the polarized $-\text{CH}_2-\text{CF}_2-$ dipoles of β -PVdF [23]. This synergy results in a highly ordered, self-anchored β -PVdF layer that is tightly integrated into the MXene framework [24]. Second, the high mechanical stiffness of MXene enables robust anchoring of the polymer chains and significantly enhances the mechanical integrity of the separator [25]. Third, the MXene surface provides continuous Li^+ conduction pathways, facilitating ion transport across the separator interface [26, 27], and mitigating the kinetic limitations typically associated with thick coating layers.

The preparation of the β -PVdF-anchored MXene scaffold (P-MX) is schematically illustrated in Figure 1d, and further details are provided in the Experimental section. The synthesis began with the selective etching of the pristine Ti_3AlC_2 MAX phase via HF treatment, which removed the Al layers to yield $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets. Subsequently, an exfoliation process was employed to achieve delamination using tetramethylammonium hydroxide (TMAOH); in this process, the intercalation of TMA^+ ions effectively expanded and separated the MXene layers, resulting in a stable colloidal dispersion of few-layer nanosheets [28].

Following delamination, the MXene nanosheets were functionalized with β -PVdF chains. PVdF was first dissolved in the high-dielectric-constant dimethyl sulfoxide (DMSO) solvent to promote spontaneous β -phase formation during continuous stirring [29]. Once the β -phase conformation was established, the delaminated MXene dispersion was introduced into the polymer solution. Strong hydrogen bonding interactions between the MXene terminations ($-\text{O}$, $-\text{OH}$, and $-\text{F}$) and the polarized $-\text{CH}_2$ groups of β -PVdF facilitated uniform anchoring of the polymer chains onto the MXene surface. The resulting P-MX composite was subsequently freeze-dried to prevent structural restacking, thereby preserving its open and porous architecture.

The structural and morphological features of the P-MX composite were elucidated through scanning electron microscopy (SEM) and transmission electron microscopy (TEM). As shown in Figure 2a and Figures S1 and S2, the etched MXene

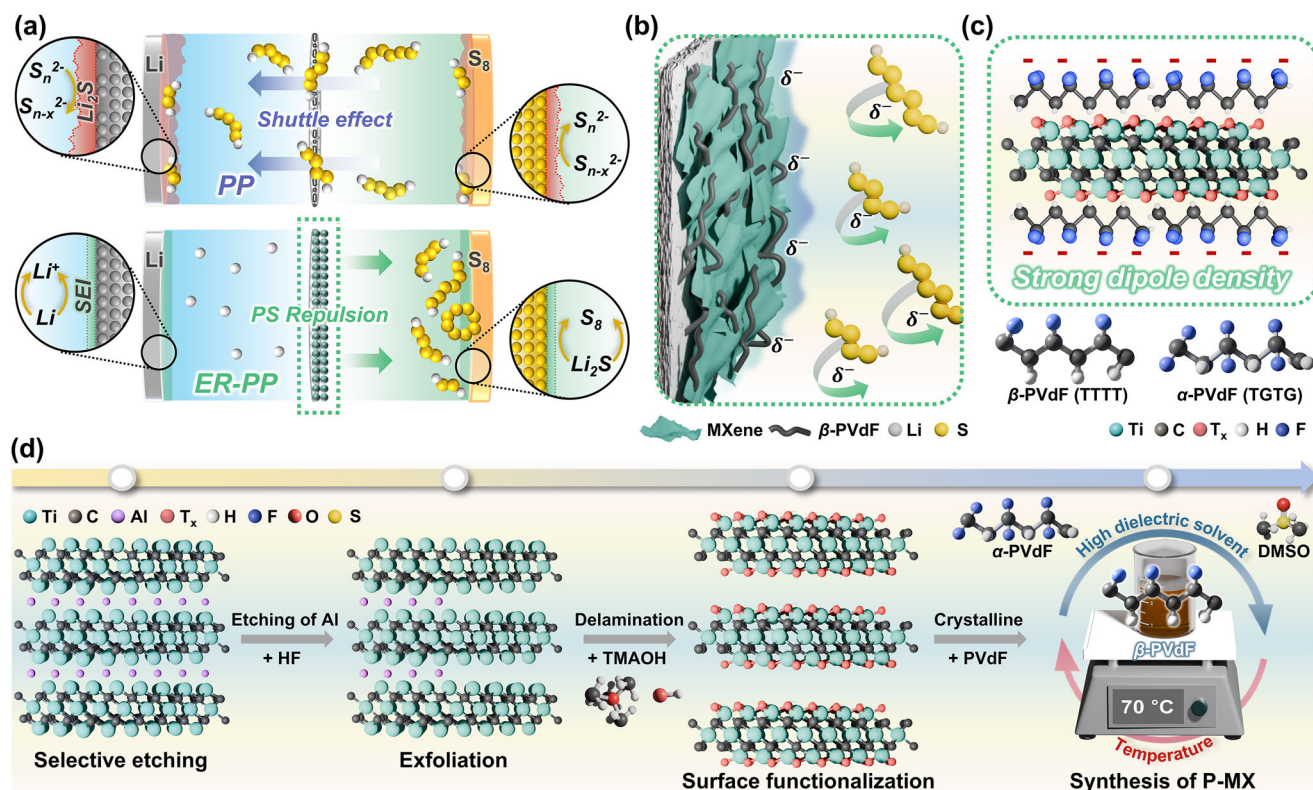


FIGURE 1 | Design and fabrication of electrostatically repulsive ER-PP separator. (a) Schematic comparison of electrochemical environments in Li-S batteries employing pristine PP and ER-PP separators. (b) Proposed mechanism for electrostatic repulsion of LiPS species mediated by the permanent dipole-induced electric field of ER-PP. (c) Schematic illustration of β-PVdF-anchored MXene scaffold (P-MX) and molecular configurations of β-PVdF (all-*trans*) and α-PVdF (*trans-gauche*). (d) Schematic workflow for the fabrication process of P-MX composite.

exhibited a significantly expanded interlayer spacing relative to its MAX precursor. This expansion originates from the selective removal of Al layers, which weakens van der Waals interactions between nanosheets and exposes abundant surface terminations [30]. Upon incorporation of the highly polar β-PVdF, the interlayer distance further increased, while the delaminated morphology of the MXene nanosheets was preserved (Figure 2b). This structural expansion is attributed to the intercalation of DMSO molecules and β-PVdF chains, which synergistically generate a more accessible and open lamellar architecture.

The TEM image of P-MX (Figure 2c) confirms that the intrinsic 2D morphology of MXene was preserved. The high-resolution TEM image acquired at the sheet edges reveals well-defined lattice fringes with an interplanar spacing of ~0.5 nm, corresponding to the (110) crystal plane [31]. Furthermore, the energy-dispersive X-ray spectroscopy (EDS) elemental mapping in Figure 2d shows that the constituent elements were uniformly distributed across the P-MX structure, without noticeable phase segregation. The homogeneously distributed fluorine signal confirms the successful incorporation of β-PVdF, suggesting that the polar polymer chains uniformly coat the MXene sheets, rather than forming localized aggregates.

To further examine the crystallographic evolution during composite synthesis, we conducted X-ray diffraction (XRD) analysis (Figure 2e). The XRD pattern clearly shows the dominant (002),

(004), and (006) reflections characteristic of MXene [32], with no detectable impurity phases. After etching, the strong (002) reflection of MXene shifted to a lower angle (7.19°), confirming the successful removal of Al layers and the formation of an expanded lamellar structure. Upon incorporation of β-PVdF, this peak exhibited a further shift to 6.55°, indicating an additional increase in interlayer *d*-spacing, which can be attributed to the steric effects and repulsive interactions introduced by the aligned dipoles. Moreover, the broadening of the (002) peak relative to pristine MXene provides direct evidence of the successful anchoring of β-PVdF chains. This broadening reflects the introduction of structural disorder and a localized distribution of interlayer spacings, resulting from the anchoring of polymer chains onto the MXene surfaces.

The increase in the interlayer spacing was further confirmed through pore structure and surface area analyses. The N₂ adsorption-desorption isotherms in Figure 2f display a clear hysteresis loop within the relative pressure (*P*/*P*₀) range of 0.45–1.0, characteristic of mesoporous structures, along with a sharp adsorption uptake near *P*/*P*₀ = 1.0, associated with macroporous voids. P-MX showed a higher adsorption capacity in this region than pristine MXene, suggesting the development of a more open and interconnected porous network. Moreover, the inset of Figure 2f reveals the presence of micropores in P-MX, confirming a hierarchical and multimodal pore size distribution. Consistently, Brunauer–Emmett–Teller (BET) analysis revealed that P-MX possessed a larger specific surface area (18.8 m² g⁻¹)

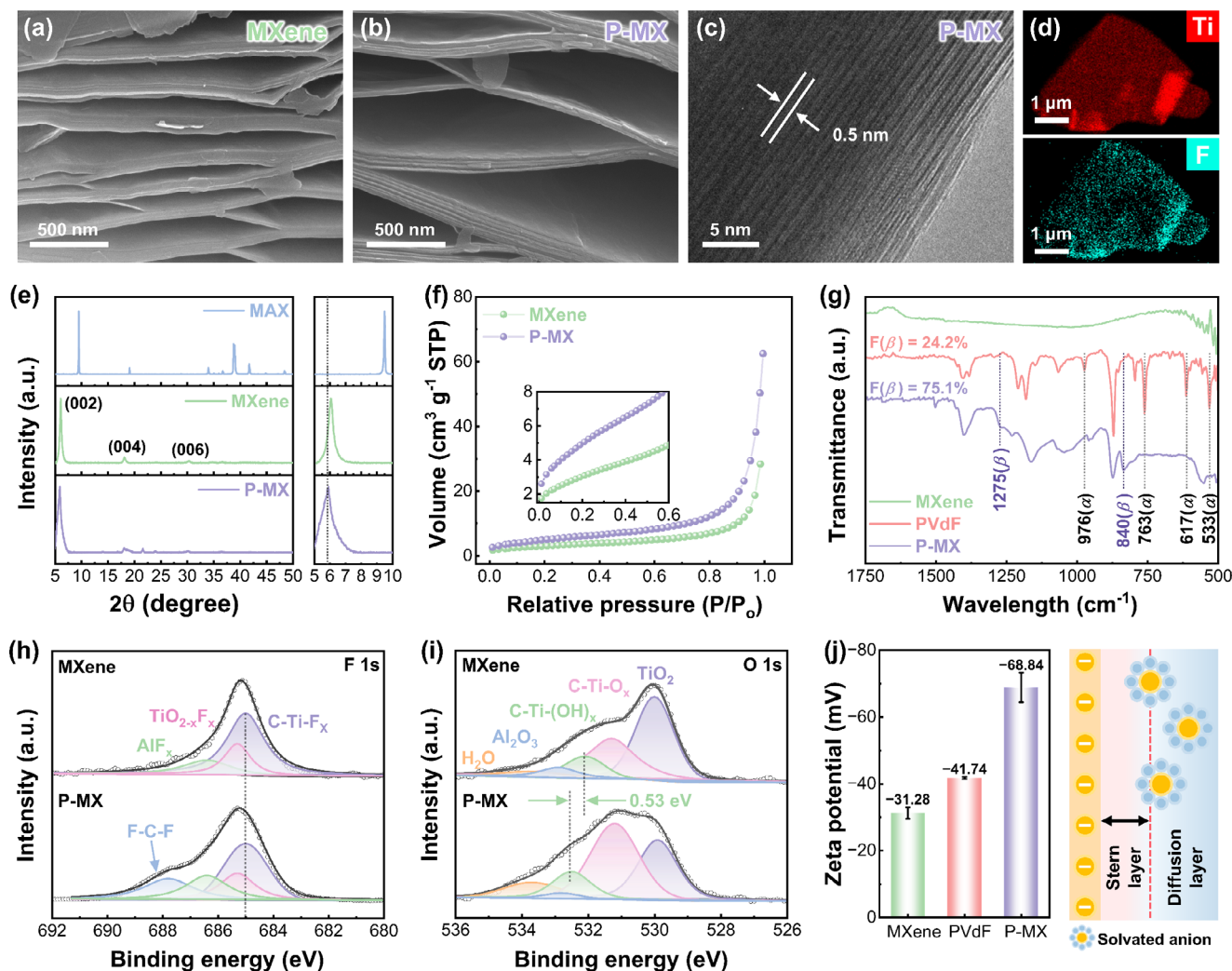


FIGURE 2 | Characterizations of MXene and P-MX composite. (a,b) SEM images showing surface morphologies of pristine MXene and P-MX, respectively. (c,d) HR-TEM image of P-MX and corresponding EDS elemental mappings for Ti (top) and F (bottom). (e) XRD patterns of MAX phase, MXene, and P-MX, with corresponding magnified regions of (002) peak. (f) N₂ adsorption-desorption isotherms for MXene and P-MX. (g) FT-IR spectra of MXene, PVdF, and P-MX. (h,i) High-resolution F 1s and O 1s XPS profiles for MXene and P-MX. (j) Zeta potential measurements (left) of MXene, PVdF, and P-MX, along with corresponding Gouy-Chapman-Stern models (right).

than MXene (9.8 m² g⁻¹). The enhanced porosity is attributed to the intercalation and anchoring of aligned β-PVdF chains between MXene sheets, which expand the interlayer spacing and prevent structural restacking.

Fourier-transform infrared (FT-IR) spectroscopy measurements were conducted to determine the chemical structure and phase transition behavior of P-MX, specifically confirming the presence of the β-phase (Figure 2g). Pristine PVdF exhibited characteristic vibrational bands corresponding to both α- and β-phases. After β-phase induction and subsequent MXene incorporation, the polar conformation was markedly strengthened and stabilized, increasing the β-phase fraction from 24.2% in pristine PVdF to 75.1% in the P-MX, as quantified based on the FT-IR spectra (see Supporting Information for the detailed calculation methods). In the P-MX spectrum, the α-phase bands at 617 and 763 cm⁻¹, as well as the γ-phase features, nearly disappeared, whereas the distinct β-phase absorption peaks at 840 cm⁻¹ became dominant, along with a weak peak at 1275 cm⁻¹ that is exclusive to the β-phase

[33]. The presence of both characteristic β-phase bands confirms the formation of β-phase PVdF in the P-MX composite. These results indicate the good alignment and stabilization of the β-PVdF chains through strong interfacial coupling with the MXene surface.

The interfacial interactions were further elucidated via X-ray photoelectron spectroscopy (XPS) analysis. The survey spectrum of P-MX (Figure S3) confirms that the material was primarily composed of Ti, C, O, and F elements. In the high-resolution F 1s spectrum (Figure 2h), MXene exhibited characteristic binding features consistent with earlier reports [34], whereas P-MX displayed a new dominant peak at 687.8 eV. This peak corresponded to the F-C-F bonding environment of aligned β-PVdF, providing direct evidence of the self-anchored features onto the MXene surfaces. In the O 1s spectrum (Figure 2i), the signal of C-Ti-OH_x groups in P-MX exhibited a positive binding energy shift of 0.53 eV relative to pristine MXene. The peak deconvolution was performed based on previously reported methodologies

(Table S1). This shift is attributed to strong hydrogen bonding interactions between the —OH terminations on MXene and the polar hydrogen-containing moieties in β -PVdF [35]. Owing to the high electronegativity of fluorine, the β -PVdF chains withdraw electron density from the MXene surface, thereby modifying its electronic environment. As a result, both the C—Ti—O_x and C—Ti bonding states shift toward higher energies, reflecting the strengthened interfacial coupling and electronic redistribution.

Having established the binding of β -PVdF chains, the key remaining question concerns the electrostatic environment generated by the aligned dipoles. Therefore, zeta-potential (ζ) measurements were performed to address this issue (Figure 2j, left). P-MX exhibited a strongly negative surface potential of −68.84 mV, indicating pronounced electrostatic repulsion of anionic LiPSs species. The magnitude of this repulsion was quantitatively evaluated based on the Gouy–Chapman–Stern (GCS) model (Figure 2j, right), with detailed assumptions and derivations provided in the Supporting Information. This analysis indicates that the surface potential of P-MX can reduce the local polysulfide concentration near the separator interface to less than ~7% of the bulk value under typical electrolyte conditions. These results confirm that the dipole-engineered interface creates a robust electrostatic barrier, effectively repelling polysulfides and suppressing their accumulation at the separator surface [36].

To evaluate the practical applicability of the P-MX composite, the prepared powder was dispersed into a slurry and integrated onto a PP separator via a simple blade coating method. The resulting P-MX-coated PP is denoted as electrostatically repulsive separator (ER-PP). For comparison, two reference separators, pristine PP and MXene-coated PP (MX-PP), were also prepared, and their intrinsic properties were evaluated (Figures S4–S6). While the pristine PP separator (Figure 3a) exhibited a smooth polyolefin surface with large, open pores characteristic of commercial separators, the surface of ER-PP was uniformly covered by interconnected microscale nanosheets (Figure 3b). The black appearance of the inset image confirms the homogeneous coverage, while the self-supporting nature of the separator was maintained; this indicates that the coating process did not compromise the mechanical integrity of the original PP separator. The cross-sectional SEM images in Figure 3c reveal that the P-MX interlayer formed a conformal coating approximately 17 μ m thick, which closely adhered to the PP framework without interfacial delamination. The corresponding EDS elemental mapping (Figure 3d) further shows that Ti and F species originating from MXene and β -PVdF, respectively, were uniformly distributed throughout the coated layer.

Sufficient porosity, superior wettability, and high electrolyte uptake are essential requirements for optimal electrolyte retention and fast Li⁺ transfer. Wettability is a critical property for separators, particularly for surface-modified ones; therefore, the electrolyte affinity of the present systems was evaluated through contact angle measurements and electrolyte uptake tests. In the wettability test (Figure 3e), pristine PP maintained a high contact angle of 30.5° after 6 s, reflecting its intrinsically hydrophobic nature with ether-based electrolytes [37]. In contrast, MX-PP showed a markedly reduced contact angle of 6.1°, attributed to the abundant polar surface terminations on MXene [38]. Notably, ER-PP exhibited complete electrolyte infiltration within 6 s, reaching

a near-zero contact angle. This exceptional wettability stems from the synergistic contribution of polar terminations of MXene and the β -PVdF chains anchored within the coating layer. Additional tests were performed to quantify the electrolyte uptake of the three separators (Figure 3f), by immersing them in the electrolyte for a fixed duration (30 s), followed by gravimetric analysis based on the mass difference before and after soaking. ER-PP exhibited a significantly higher electrolyte uptake (95.1%) compared to the pristine PP and MX-PP. The enhanced surface polarity and dielectric environment of ER-PP significantly improve its electrolyte affinity, ensuring rapid wetting and facilitating the formation of continuous Li⁺ transport pathways.

To assess how the enhanced electrolyte wettability and uptake affect the interfacial electrochemical kinetics, electrochemical impedance spectroscopy (EIS) measurements were performed prior to cycling (Figure 3g). The bulk resistance (R_s) values of PP, MX-PP, and ER-PP were 2.90, 3.85, and 3.92 Ω , respectively. The slightly higher R_s observed for the MX-PP and ER-PP separators can be attributed to the introduction of the coating layer, which slightly increases the effective thickness and tortuosity of the ion transport pathway across the separator. Nevertheless, the absolute R_s values remained low, indicating that the coating did not severely hinder bulk ionic conduction. Similarly, the three separators exhibited comparable charge-transfer resistance (R_{ct}) values of 11.87, 6.74, and 8.30 Ω for PP, MX-PP, and ER-PP, confirming that the incorporation of the dipole-engineered β -PVdF layer did not significantly compromise the interfacial kinetics, while enabling the desired electrostatic repulsion functionality.

Besides the electrolyte affinity, further tests were conducted to investigate the mechanical and thermal robustness of the separators. The mechanical flexibility of the ER-PP separator was evaluated through repetitive bending tests (Figure 3h). The ER-PP separator maintained excellent flexibility and structural integrity; repeated bending induced no cracking or interfacial peeling, indicating strong adhesion and effective stress transfer between the P-MX coating and the PP substrate. The thermal stability of the samples was further assessed by monitoring dimensional changes at elevated temperatures (Figure 3i). Pristine PP underwent severe thermal shrinkage, retaining only 48.6% of its initial area at 160°C. In contrast, MX-PP exhibited markedly improved thermal stability, maintaining 89.8% of its original size, owing to the intrinsic thermal tolerance and heat-dissipating capability of MXene [39]. Notably, ER-PP exhibited superior thermal stability, showing negligible dimensional changes up to 100°C and retaining 93.9% of its initial size even at 160°C. These results show that the combination of MXene and β -PVdF produces a mechanically stronger and thermally resilient framework.

Furthermore, tensile tests (Figure 3j) revealed that the ER-PP separator exhibited a high tensile strength of 82.6 MPa, along with an elongation at break of 24.6%. These results indicate that ER-PP significantly outperformed pristine PP in terms of mechanical strength and flexibility, while exhibiting an improved tensile strength compared to the MX-PP separator (68.2 MPa). This improvement is attributed to the PVdF chains acting as structural binders [40], which tightly anchor adjacent MXene sheets and enhance the integrity of the interlayer, enabling efficient stress transfer under tensile deformation.

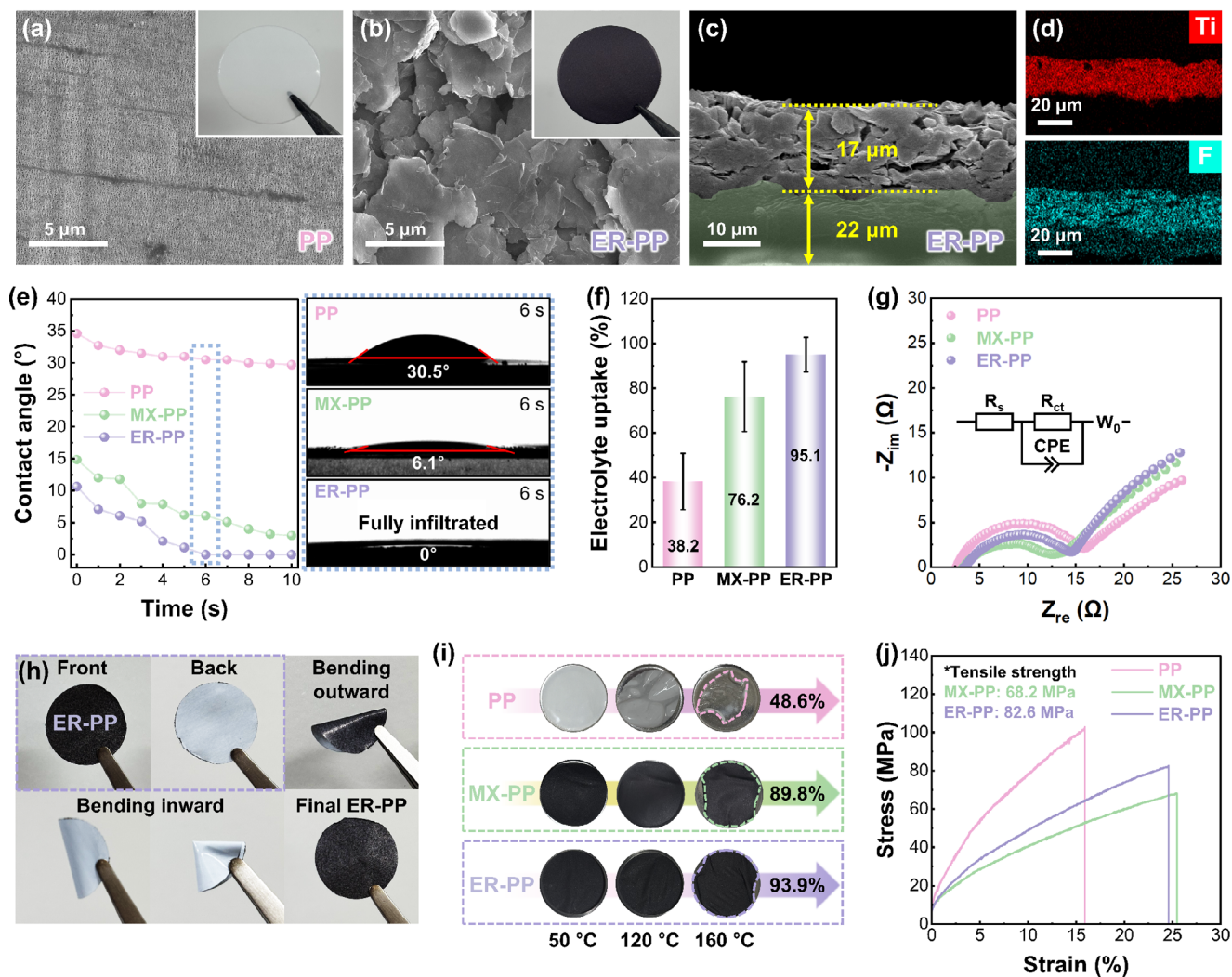


FIGURE 3 | Physical, mechanical, and electrolyte affinity characterization of ER-PP separators. (a,b) Top-view SEM images of PP and ER-PP separators, along with corresponding optical photographs (insets). (c,d) Cross-sectional SEM image of ER-PP and corresponding EDS elemental mappings of Ti (top) and F (bottom). (e) Time evolution of electrolyte contact angles over 10 s (left), along with contact angles at 6 s (right). (f) Quantitative electrolyte uptake of PP, MX-PP, and ER-PP separators. (g) EIS Nyquist plots and corresponding equivalent circuit fittings of PP, MX-PP, and ER-PP cells prior to cycling. (h) Bending and handling tests of ER-PP separator. (i) Thermal shrinkage behavior at elevated temperature and (j) tensile stress-strain curves of PP, MX-PP, and ER-PP separators.

Most importantly, as the primary function of the proposed ER-PP is associated with its polysulfide-repulsive interface, it is essential to directly evaluate its repulsion capability. To this end, LiPS permeation tests were conducted using an H-type diffusion cell (Figure 4a–c). The left chamber was filled with 0.05 M Li_2S_6 dissolved in 1,3-dioxolane/1,2-dimethoxyethane (DOL/DME) (1:1 v/v, 1 M LiTFSI (lithium bis(trifluoromethanesulfonyl)imide)), while the right chamber contained only the blank solvent. With the pristine PP separator, the right chamber began to turn yellow almost immediately, indicating the rapid diffusion of LiPSs (Figure 4a). After 24 h, the entire chamber became dark yellow, consistent with the nonpolar and non-interactive nature of PP. The MX-PP separator exhibited noticeably slower color evolution (Figure 4b), owing to the polar surface terminations that partially delayed the LiPSs crossover via weak adsorption; however, the gradual color intensification in the right chamber indicates that the crossover suppression was ultimately insufficient. In sharp contrast, no visible color change in the right chamber was

observed for the ER-PP separator throughout the diffusion period (Figure 4c). This markedly enhanced suppression demonstrates that the strongly aligned C–F dipoles establish a stable, negatively polarized interface, which electrostatically repels S_x^{2-} species and prevents them from approaching the separator. This effective push-back mechanism minimizes LiPSs accumulation within the separator, thereby establishing a robust electrostatic gatekeeping function.

To quantitatively confirm these differences, we conducted UV-vis and Raman spectroscopy experiments. The solution collected from the right chamber (i.e., the permeated side) was diluted and analyzed by UV-vis spectroscopy (Figure 4d). As the UV absorbance in the 250–300 nm range is directly proportional to the concentration of Li_2S_6 , the relative absorbance intensities provide a quantitative measure of the LiPSs crossover. The PP separator exhibited the highest absorbance, whereas MX-PP showed a moderately reduced absorbance. Notably, ER-PP

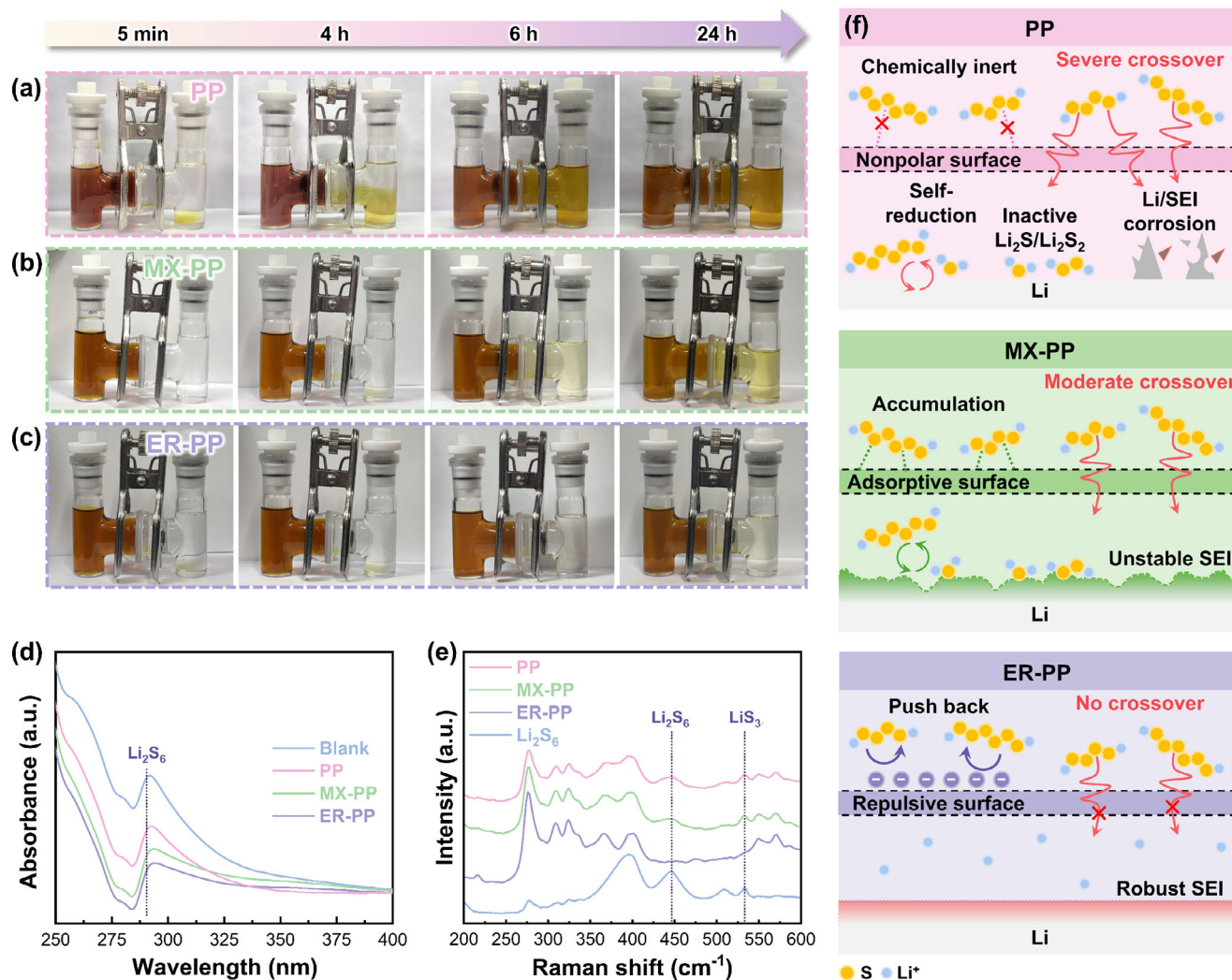


FIGURE 4 | Evaluation of polysulfide permeation and transport mechanisms. Time-dependent optical photographs of H-type diffusion cells monitored over 24 h to compare the LiPSs-blocking capabilities of (a) PP, (b) MX-PP, and (c) ER-PP separators. (d) UV-vis spectra and (e) Raman spectra of solutions collected from the right chamber (permeate side) after 24 h. (f) Schematic illustration comparing LiPSs transport mechanisms of PP, MX-PP, and ER-PP separators in Li-S batteries.

yielded the lowest absorbance among all samples, demonstrating an effective push-back effect. Consistently, the Raman spectra in Figure 4e revealed distinct LiPSs-related peaks at 445 cm^{-1} (S_6^{2-}) and 533 cm^{-1} (S_3^{2-}) for the PP and MX-PP separators, indicating insufficient suppression of polysulfide crossover. In sharp contrast, only negligible polysulfide signals were detected for the ER-PP separator, confirming its superior ability to inhibit LiPSs migration.

To clarify the microscopic origin of the electrostatic gatekeeping effect, density functional theory (DFT) calculations were carried out for P-MX with different β -PVdF molecular adsorption orientations (Figures S7 and S8, Table S2). The results reveal that β -PVdF induces a strongly polarized interface (6.02 D) than MXene (0.008 D). Furthermore, the P-MX surface becomes thermodynamically unfavorable for LiPSs adsorption (Figure S8), indicating that the aligned β -PVdF layer transforms MXene from an intrinsically adsorptive surface into an electrostatic gatekeeper that suppresses LiPSs adsorption.

This polysulfide migration trend (PP > MX-PP > ER-PP) directly reflects the distinct transport mechanisms of each separator (Figure 4f). The nonpolar PP separator enables free LiPSs crossover, accelerating parasitic reduction reactions at the anode side and leading to severe Li/solid electrolyte interphase (SEI) interface degradation. Although MX-PP partially delays LiPSs migration through its polar surface terminations, it still suffers from moderate crossover. In contrast, ER-PP effectively suppresses LiPSs migration via strong electrostatic repulsion, thereby minimizing active material loss and mitigating the shuttle effect. This repulsion-governed mechanism establishes a more stable interfacial environment for the Li-metal anode, leading to improved electrochemical performance.

To investigate the contribution of the ER-PP separator, three types of separators were assembled into Li-S cells, and their electrochemical performances were compared. Cyclic voltammetry (CV) measurements were conducted at a scan rate of 0.1 mV s^{-1} , under conditions where polysulfides were already

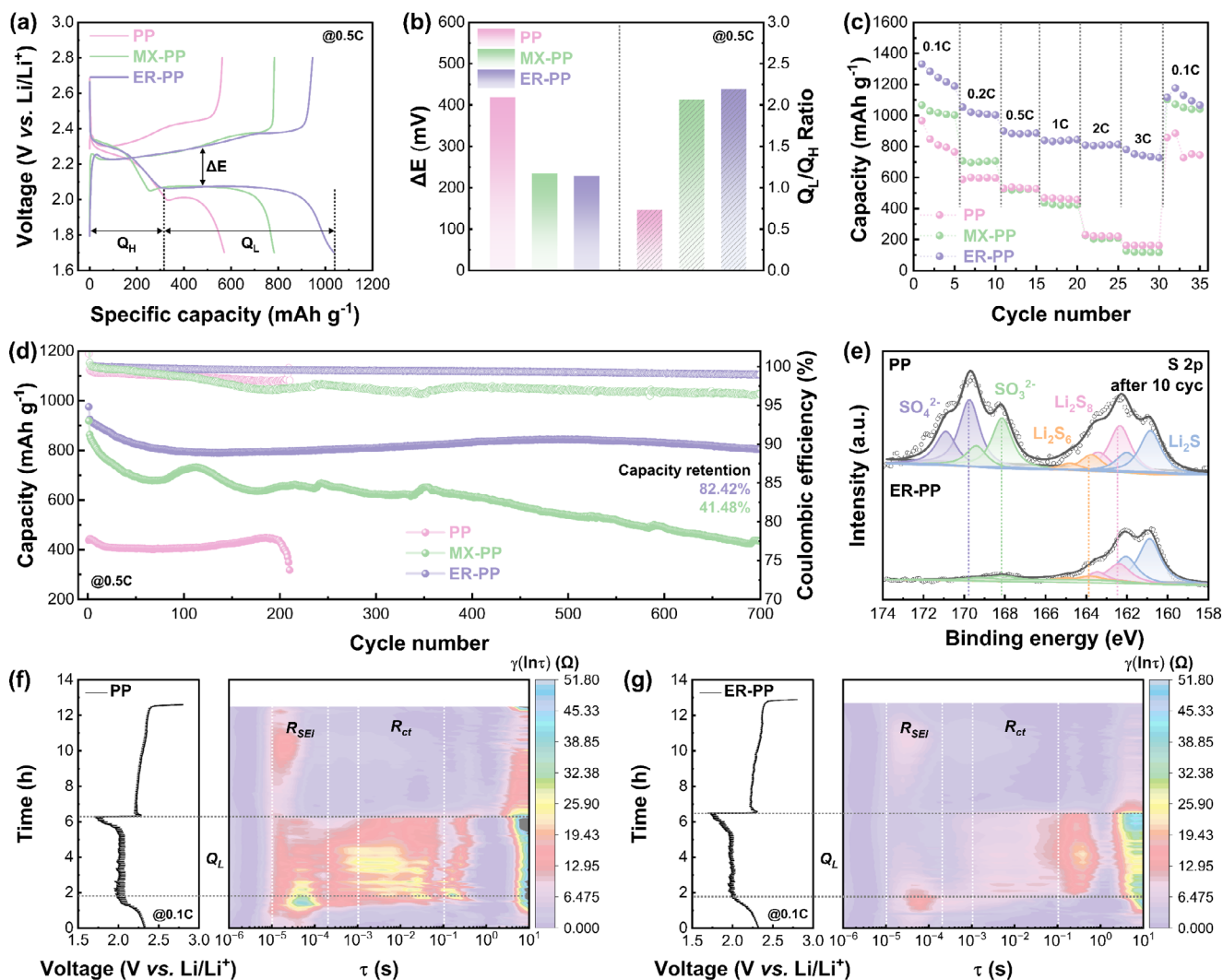


FIGURE 5 | Electrochemical performance of Li-S cells. (a) GCD profiles obtained using different separators at 0.5C. (b) ΔE and Q_L/Q_H ratios obtained from GCD profiles. (c) Rate capabilities of PP, MX-PP, and ER-PP cells at current densities ranging from 0.1 to 3C. (d) Long-term cycling performance and Coulombic efficiency of PP, MX-PP, and ER-PP cells at 0.5C (1C = 1675 mAh g⁻¹). (e) High-resolution XPS spectra of Li-metal surfaces after 10 cycles with the PP and ER-PP separators. In situ DRT analysis of the EIS measurements in Li-S batteries with (f) PP and (g) ER-PP separators.

dissolved and actively participating in the redox processes (Figure S9). Notably, the ER-PP cell exhibited a higher peak current and narrower polarization gap (300.9 mV) compared to those of the pristine PP (375.2 mV) and MX-PP (362.0 mV) cells. This reduced polarization indicates accelerated redox kinetics and improved reversibility [41]. Such enhancement is attributed to the stable polysulfide-repulsive surface, which continuously redirects free polysulfides to the cathode region, thereby maintaining an active redox environment and enhancing sulfur utilization. In addition, the reduced polarization indicates a stabilized Li-metal interface, as the limited polysulfide migration effectively prevents parasitic corrosion reactions at the Li anode.

These results are further supported by the comparison of initial galvanostatic charge/discharge (GCD) profiles (Figure 5a). All three cells exhibited a well-defined two-step conversion behavior between S₈ and Li₂S, consistent with the CV results. The cell employing the ER-PP separator delivered longer discharge/charge plateaus with smaller potential gaps compared to the counterparts, as summarized in Figure 5b. The higher

low-to-high plateau capacity ratio (Q_L/Q_H) of the ER-PP cell emerges as a more favorable conversion of LiPSs to short-chain sulfides [42].

Owing to the LiPSs-blocking effect, the ER-PP separator significantly improves both the redox kinetics and cycling stability. Rate performance tests (Figure 5c) revealed that, while the pristine PP cell delivered an initial discharge capacity of 967 mAh g⁻¹ at 0.1C, its capacity decreased sharply to only 161 mAh g⁻¹ at 3C. In sharp contrast, the cell employing the ER-PP separator delivered a high initial discharge capacity of 1331 mAh g⁻¹, which remained at 781 mAh g⁻¹ even at 3C. Importantly, when the current density was switched from 3C back to 0.1C, the cell recovered a high discharge capacity of 1116 mAh g⁻¹, demonstrating its excellent rate reversibility.

Besides the improved rate capability, the ER-PP separator also provides remarkable long-term cycling stability. The cycling performance of Li-S cells was evaluated at 0.5C (Figure 5d). The cell with the PP separator exhibited poor stability, undergoing

rapid failure after 200 cycles, while the MX-PP cell showed relatively low-capacity retention (41.5% after 700 cycles). In sharp contrast, the cell with the ER-PP separator maintained a highly stable long-term cycling behavior, delivering a reversible capacity of 804 mAh g⁻¹ after 700 cycles, with 82.4% capacity retention. Furthermore, the ER-PP-based cell demonstrated exceptional long-term cycling stability even under a high-rate condition of 3C (Figure S10), maintaining stable operation for over 1400 cycles with a high Coulombic efficiency of 98.68%. To evaluate its practical viability, we fabricated Li-S pouch cells (Figure S11), where the ER-PP separator significantly outperformed the conventional PP by effectively mitigating the shuttle effect and maintaining a high capacity of 814.8 mAh g⁻¹ after 40 cycles.

To gain insight into the kinetic mechanism of the Li-S cells, their real-time internal resistance and ion transport behavior were analyzed using the galvanostatic intermittent titration technique (GITT, Figure S12). Consistent with above electrochemical results, the ER-PP cell exhibited the lowest internal resistance during both the charge and discharge cycles. This enhanced kinetic behavior is further supported by a higher Li⁺ diffusion coefficient (D_{Li^+}) of 1.58×10^{-8} cm² s⁻¹ for ER-PP, which represents an order-of-magnitude improvement compared with those of pristine PP (1.01×10^{-9} cm² s⁻¹) and MX-PP (4.92×10^{-9} cm² s⁻¹). By effectively preventing LiPSs accumulation at the separator interface, the ER-PP separator reduces unfavorable ion-polysulfide interactions that typically impede Li⁺ migration [43], thereby improving the reaction kinetics and enhancing the overall cell stability.

The extent of polysulfide-induced corrosion on the Li anode was investigated using XPS and ex situ SEM. In the S 2p spectrum (Figure 5e), the Li anode cycled with the PP separator exhibited pronounced peaks at 162.4 and 163.8 eV, corresponding to long-chain LiPSs species, together with prominent peaks at 168.1 and 169.8 eV associated with LiSO_x formation [44]. During cycling, active Li is continuously exposed to and reacts with crossover LiPSs, owing to the repeated breakdown of SEI. These parasitic reaction products (LiSO_x) not only increase anode polarization, but also lead to irreversible capacity loss, thereby severely degrading the overall electrochemical performance [45]. Remarkably, the integration of ER-PP effectively suppresses the side reactions by limiting polysulfide access to the Li surface, resulting in negligible detection of LiSO_x-related peaks.

Consistent with the XPS results, the ex situ SEM images of the cycled Li anode surfaces reveal pronounced morphological differences (Figure S13). In cells with the pristine PP separator, the Li surface was characterized by numerous pits, originating from LiPSs-induced corrosion. Although the MX-PP separator provided a marginal reduction in dendritic growth compared to the pristine PP, the Li surface still exhibited significant roughness and localized cracking, indicating that a simple MXene coating is insufficient to completely prevent polysulfide-mediated interfacial degradation. In addition, extensive surface cracking was observed in these cells, presumably caused by internal stress arising from uneven lithiation/delithiation, a primary driver of gradual capacity decay [46]. On the other hand, the Li anode coupled with the ER-PP separator maintained a dense and smooth morphology. In addition, ex situ SEM analysis of the separator after cycling revealed no noticeable structural

deformation, further indicating the robust interfacial stability of the P-MX layer (Figure S14). These observations confirm that the strong electrostatic repulsion can effectively reduce polysulfides crossover and suppress Li corrosion, thereby contributing to enhanced sulfur utilization and internal cell stability.

While the XPS and ex situ SEM analyses clearly demonstrate the suppressed polysulfide-induced corrosion and improved interfacial stability at the Li anode, a deeper understanding of how these interfacial characteristics influence the overall electrochemical kinetics is still required. To this end, we performed an in situ EIS-Distribution of Relaxation Times (DRT) analysis, which provides superior resolution in deconvoluting overlapping electrochemical processes compared to traditional Nyquist plots (Figure 5f,g). The peaks in the high-frequency range ($\tau = 10^{-5} - 10^{-4}$ s), corresponding to R_{SEI} , are primarily associated with the ion transport through the SEI and interfacial resistances at both the anode and cathode. Notably, the ER-PP cell exhibits a significant reduction in these resistance intensities compared to the PP cells. This reduction is attributed to the negatively polarized interface of the ER-PP separator, which effectively repels polysulfide anions and suppresses their parasitic reactions at the Li-metal surface, leading to the formation of a more stable and thinner SEI layer.

Furthermore, the peaks in the mid-to-low frequency range ($\tau = 10^{-3} - 10^{-1}$ s) correspond to the R_{ct} and the conversion kinetics of polysulfide species at the cathode. The ER-PP cell shows a marked decrease in the R_{ct} , indicating accelerated sulfur redox kinetics. This suggests that the electrostatic confinement of polysulfides within the cathode region maintains a high concentration of active species, thereby facilitating more efficient conversion reactions. These DRT findings provide definitive evidence that the ER-PP separator simultaneously optimizes the anode interfacial stability and cathode reaction kinetics by effectively regulating polysulfide transport.

Taken together, these multiple electrochemical and interfacial analyses underscore that the polarized interface of the ER-PP separator provides a fundamental solution to address the long-standing challenges associated with both the sulfur cathode and the Li anode.

3 | Conclusion

In summary, we have shown that a separator coated with aligned β -PVdF-anchored MXene (ER-PP) provides an effective polysulfide-repulsive barrier for Li-S batteries. By exploiting highly oriented C-F dipoles, we established a persistent electrostatic field that actively repels anionic S_x²⁻ species, thereby maintaining a pristine separator environment even under prolonged cycling. Distinct from conventional adsorption-based strategies, this dipole-induced repulsive mechanism reported here actively confines LiPSs species within the cathode region and successfully suppresses the shuttle effect, thereby enhancing the Li⁺ transport and redox kinetics. As a result, the ER-PP cells exhibit a markedly improved cycling durability, retaining a high reversible capacity of 803.6 mAh g⁻¹ after 700 cycles. This work overcomes the limitations of traditional trap-and-hold approaches by introducing a robust, interfacial polarization-driven repulsive paradigm. Furthermore, the facile and scalable

blade coating process highlights the practical viability of the ER-PP separator for large-scale energy storage applications.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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